Лабораторная работа №1

Моделирование сетей передачи данных

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Цель лабораторной работы

- Развернуть в системе виртуализации (например, в VirtualBox) mininet, познакомиться с основными командами для работы с Mininet через командную строку и через графический интерфейс.

Выполнение лабораторной работы

Настройка образа VirtualBox

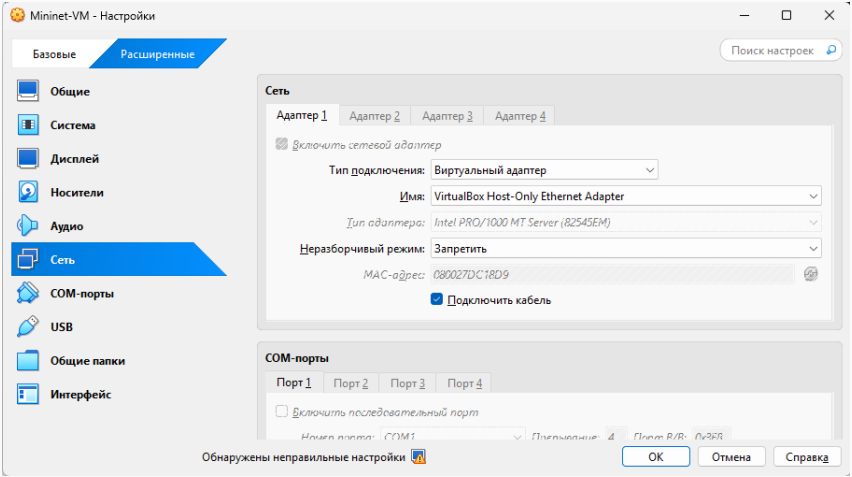

Figure 1: Установка и настройка виртуальной машины

Подключение к виртуальной машине

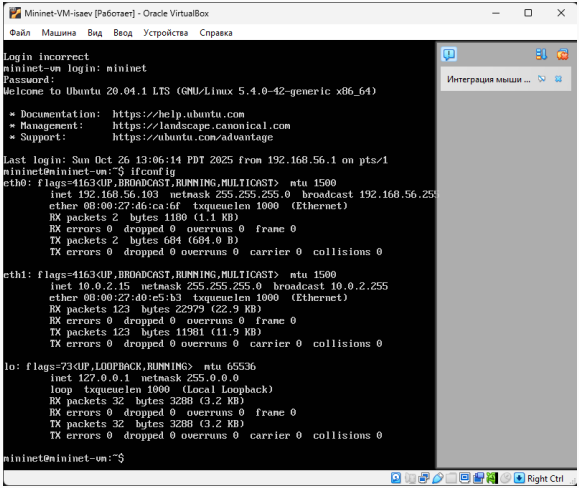

Figure 2: Вход и просмотр адреса виртуальной машины

Подключение к виртуальной машине

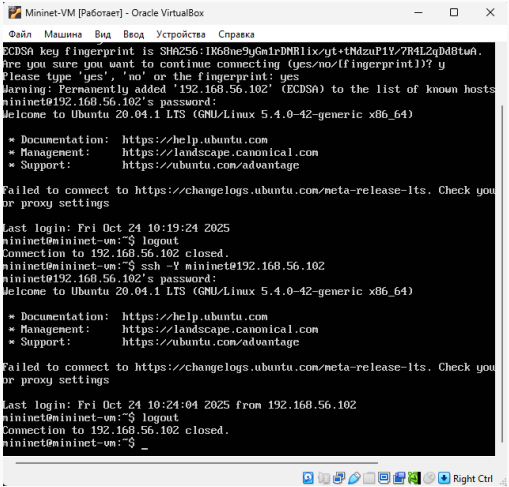

Figure 3: Подключение к виртуальной машине из терминала хостовой машины

Работа с Mininet из-под Windows

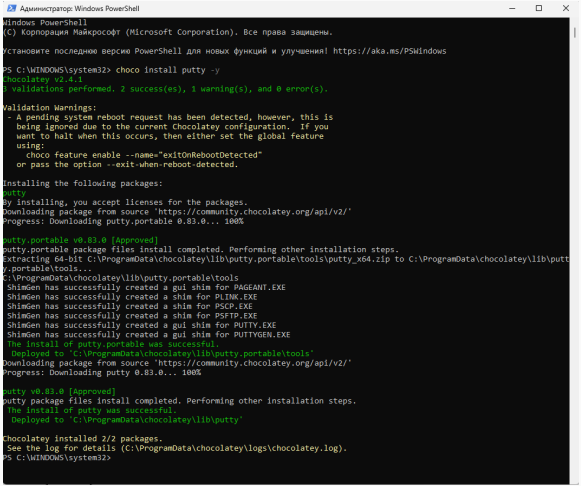

Figure 4: Установка putty

Работа с Mininet из-под Windows

Администратор: Windows PowerShell

```
 See the log for details (C:\ProgramData\chocolatey\logs\chocolatey.log).
PS C:\WINDOWS\system32> choco install vcxsrv -y
Chocolatey v2.4.1
3 validations performed. 2 success(es), 1 warning(s), and 0 error(s).

Validation Warnings:
 - A pending system reboot request has been detected, however, this is
   being ignored due to the current Chocolatey configuration.  If you
   want to halt when this occurs, then either set the global feature
   using:
     choco feature enable --name="exitOnRebootDetected"
   or pass the option --exit-when-reboot-detected.

Installing the following packages:
vcxsrv
By installing, you accept licenses for the packages.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading vcxsrv 21.1.10... 100%

vcxsrv v21.1.10 [Approved]
vcxsrv package files install completed. Performing other installation steps.
Installing 64-bit vcxsrv...
vcxsrv has been installed.
  vcxsrv may be able to be automatically uninstalled.
 The install of vcxsrv was successful.
  Software installed as 'EXE', install location is likely default.

Chocolatey installed 1/1 packages.
 See the log for details (C:\ProgramData\chocolatey\logs\chocolatey.log).
PS C:\WINDOWS\system32>
```


Figure 5: Установка putty VcXsrv Windows X Server

Работа с Mininet из-под Windows

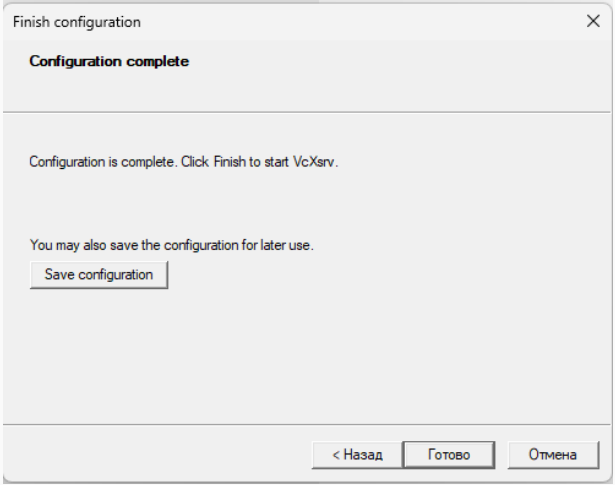

Figure 6: Запуск и настройка Xserver

Работа с Mininet из-под Windows

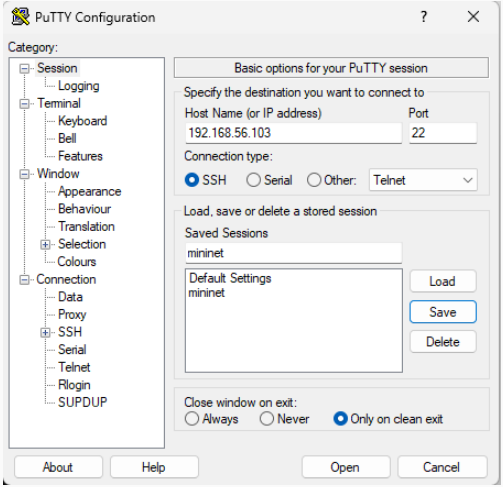

Figure 7: Запуск putty и добавление опции перенаправления X11

Настройка параметров XTerm

```
mininet@mininet-vm: ~
/etc/X11/app-defaults/XTerm   [----] 18 L:[239+27 266/266] *(10376/10376b) <EOF>                    [*][X]
! sizes.  Uncomment this line to use the same size as "fixed".
!*faceSize: 8

! Here is a pattern that is useful for double-clicking on a URL:
*charClass: 33:48,35:48,37-38:48,43-47:48,58:48,61:48,63-64:48,95:48,126:48
!
! Alternatively,
!*on2Clicks: regex [[:alpha:]]+://([[:alnum:]!#+,./=?@_~-]|(%[[:xdigit:]][[:xdigit:]]))+

! VT100s and similar terminals recognize escape sequences and control
! characters to which they reply to the host with other escape sequences,
! to provide information.  The "resize" program uses this feature.
!
! In addition, xterm recognizes several escape sequences which can be used to
! set fonts, window properties, return settings via escape sequences.  Some
! find these useful; others are concerned with the possibility of unexpected
! inputs.
!
! All of these features can be enabled or disabled via menus.
!
! Depending on your environment, you may wish to disable those by default by
! uncommenting one or more of the resource settings below:
!*allowFontOps: false
!*allowTcapOps: false
!*allowTitleOps: false
!*allowWindowOps: false
xterm*faceName: Monospace
xterm*faceSize: 12
 1Help    2Save    3Mark    4Replac  5Copy    6Move    7Search  8Delete  9PullDn  10Quit
```

Figure 8: Увелечение размера шрифта и применение векторного шрифта

Настройка соединения X11 для суперпользователя

```
mininet@mininet-vm:~$ xauth list $DISPLAY
mininet-vm/unix:10  MIT-MAGIC-COOKIE-1  69fa6826576af937fcefe92dc91d92ad
mininet@mininet-vm:~$ sudo -i
root@mininet-vm:~# xauth list
mininet-vm/unix:10  MIT-MAGIC-COOKIE-1  69fa6826576af937fcefe92dc91d92ad
root@mininet-vm:~#
```

Figure 9: Заполнения файла полномочий /root/ .Xauthority

Работа с Mininet с помощью командной строки

root@mininet-vm: ~

```
mininet-vm/unix:10  MIT-MAGIC-COOKIE-1  69fa6826576af937fcefe92dc91d92ad
root@mininet-vm:~# sudo mn
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet>
```

Figure 10: Вызов Mininet с использованием топологии по умолчанию

Работа с Mininet с помощью командной строки

```
mininet> help

Documented commands (type help <topic>):
========================================
EOF    gterm  iperfudp  nodes        pingpair      py      switch  xterm
dpctl  help   link      noecho       pingpairfull  quit    time
dump   intfs  links     pingall      ports         sh      wait
exit   iperf  net       pingallfull  px            source  x

You may also send a command to a node using:
  <node> command {args}
For example:
  mininet> h1 ifconfig

The interpreter automatically substitutes IP addresses
for node names when a node is the first arg, so commands
like
  mininet> h2 ping h3
should work.

Some character-oriented interactive commands require
noecho:
  mininet> noecho h2 vi foo.py
However, starting up an xterm/gterm is generally better:
  mininet> xterm h2

mininet>
```

Figure 11: Отображение списка команд и примеров их использования

```
mininet> nodes
available nodes are:
c0 h1 h2 s1
mininet>
```

Figure 12: Отображение доступных узлов

```
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
s1 lo:  s1-eth1:h1-eth0 s1-eth2:h2-eth0
c0
mininet>
```

Figure 13: Просмотр доступных линков

Работа с Mininet с помощью командной строки

```
mininet> h1 ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.0.1  netmask 255.0.0.0  broadcast 10.255.255.255
        ether ca:29:e3:72:ab:3b  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

mininet>
```

Figure 14: Выполнение команды для устройства h1

Работа с Mininet с помощью командной строки

```
mininet> h1 ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.329 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.128 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.121 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.102 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.082 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.093 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.403 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.098 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.088 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.229 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.305 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.086 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.073 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.128 ms
^C
--- 10.0.0.2 ping statistics ---
15 packets transmitted, 15 received, 0% packet loss, time 14319ms
rtt min/avg/max/mdev = 0.061/0.155/0.403/0.104 ms
mininet> exit
*** Stopping 1 controllers
c0
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
completed in 185.246 seconds
root@mininet-vm:~#
```

Figure 15: Проверка связи между узлами h1 и h2

Работа с Mininet с помощью командной строки

```
root@mininet-vm:~# sudo mn -c
*** Removing excess controllers/ofprotocols/ofdatapaths/pings/noxes
killall controller ofprotocol ofdatapath ping nox_core lt-nox_core ovs-openflowd ovs-controller ovs-testcontroller udpbw
test mnexec ivs ryu-manager  2> /dev/null
killall -9 controller ofprotocol ofdatapath ping nox_core lt-nox_core ovs-openflowd ovs-controller ovs-testcontroller ud
pbwtest mnexec ivs ryu-manager  2> /dev/null
pkill -9 -f "sudo mnexec"
*** Removing junk from /tmp
rm -f /tmp/vconn* /tmp/vlogs* /tmp/*.out /tmp/*.log
*** Removing old X11 tunnels
*** Removing excess kernel datapaths
ps ax | egrep -o 'dp[0-9]+' | sed 's/dp/nl:/'
***  Removing OVS datapaths
ovs-vsctl --timeout=1 list-br
ovs-vsctl --timeout=1 list-br
*** Removing all links of the pattern foo-ethX
ip link show | egrep -o '([-_.[:alnum:]]+-eth[[:digit:]]+)'
ip link show
*** Killing stale mininet node processes
pkill -9 -f mininet:
*** Shutting down stale tunnels
pkill -9 -f Tunnel=Ethernet
pkill -9 -f .ssh/mn
rm -f ~/.ssh/mn/*
*** Cleanup complete.
root@mininet-vm:~#
```

Figure 16: Очистка предыдущего экземпляра Mininet

Построение и эмуляция сети в Mininet с использованием графического интерфейса

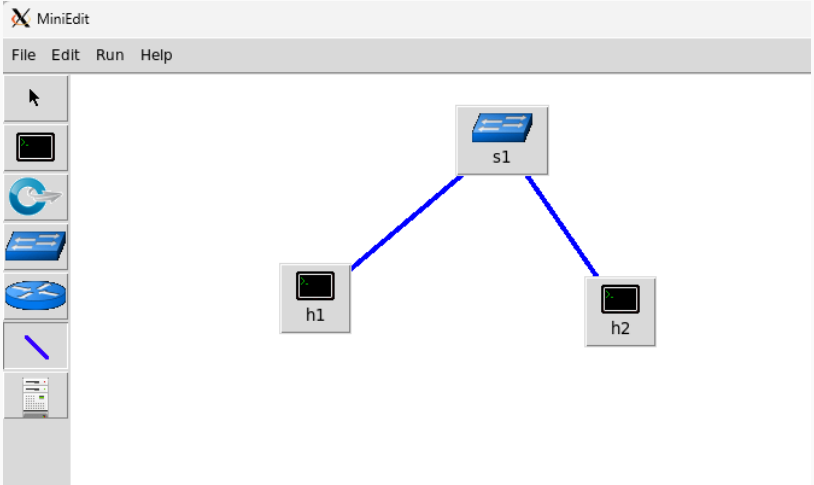

Figure 17: Добавление двух хостов и одного коммутатора

Построение и эмуляция сети в Mininet с использованием графического интерфейса

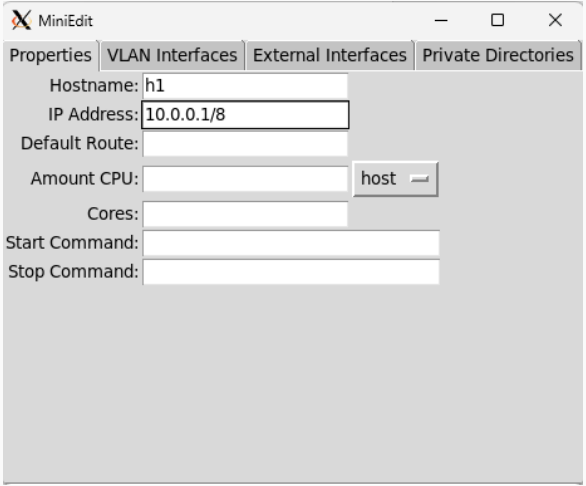

Figure 18: Настройка IP-адреса на хосте h1

Построение и эмуляция сети в Mininet с использованием графического интерфейса

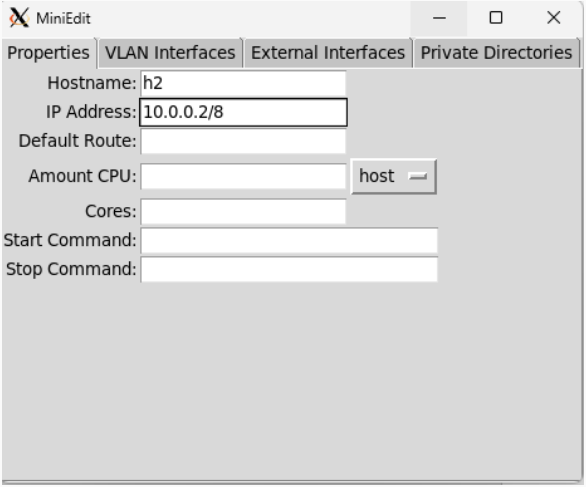

Figure 19: Настройка IP-адреса на хосте h2

Построение и эмуляция сети в Mininet с использованием графического интерфейса

"Host: h2"@mininet-vm

```
root@mininet-vm:/home/mininet# ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.0.2  netmask 255.0.0.0  broadcast 10.255.255.255
        ether 96:72:ec:c9:e6:b4  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 1937  bytes 477076 (477.0 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 1937  bytes 477076 (477.0 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@mininet-vm:/home/mininet#
```

Figure 20: Проверка назначенных IP-адресов для h2 и проверка соединения между хостами

Построение и эмуляция сети в Mininet с использованием графического интерфейса

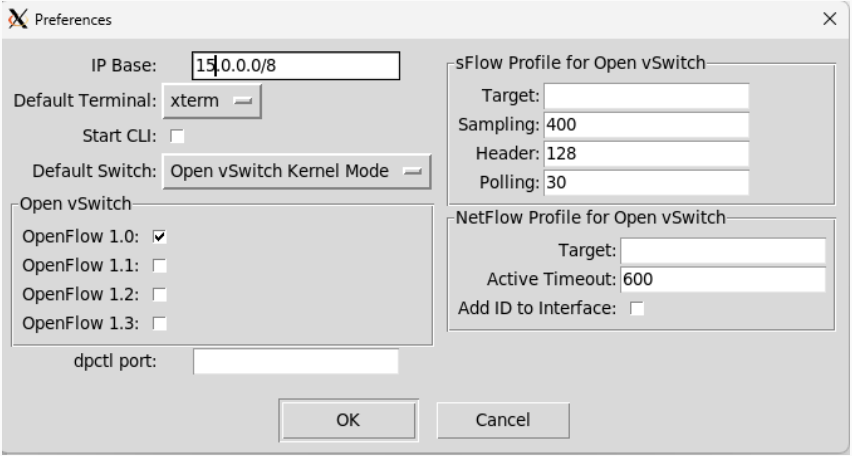

Figure 21: Проверка автоматического назначения адресов

"Host: h1"@mininet-vm

```
root@mininet-vm:/home/mininet# ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.0.1  netmask 255.0.0.0  broadcast 10.255.255.255
        ether 82:b5:d4:9f:08:c9  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 3023  bytes 424900 (424.9 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 3023  bytes 424900 (424.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@mininet-vm:/home/mininet#
```

Figure 22: Отображение IP-адреса, назначенного хосту h1

```
mininet@mininet-vm:~$ mkdir ~/work
mkdir: cannot create directory '/home/mininet/work': File exists
mininet@mininet-vm:~$ ls
mininet  mininet.orig  oflops  oftest  openflow  pox  work
mininet@mininet-vm:~$
```

Figure 23: Создание нового каталога

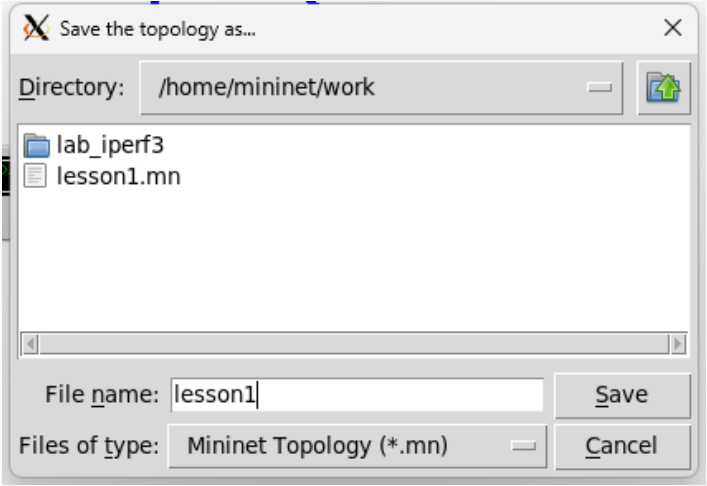

Figure 24: Сохранение топологии

```
mininet@mininet-vm:~/work$ ls
lab_iperf3  lesson1.mn
mininet@mininet-vm:~/work$
```

Figure 25: Изменение прав доступа к файлам в каталоге проекта

Вывод

- В ходе выполнения лабораторной работы были получены навыки по развёртываню в системе виртуализации (например, в VirtualBox) mininet, а также познакомились с основными командами для работы с Mininet через командную строку и через графический интерфейс.